

Steering Vision Models towards Subjective Concepts with Language

Mihret Agegnehu Bekele *
mihretagegnehu2@gmail.com
Addis Ababa Science and Technology
University
Addis Ababa, Ethiopia

Dawit Getahun Mengistu*
dawit.getahun-ug@aau.edu.et
Addis Ababa University
Addis Ababa, Ethiopia

Kidus Paulos
pauloskidus44@gmail.com
Saint Mary's University
Addis Ababa, Ethiopia

Natnael Abayneh
natnael.abayneh-ug@aau.edu.et
Addis Ababa University
Addis Ababa, Ethiopia

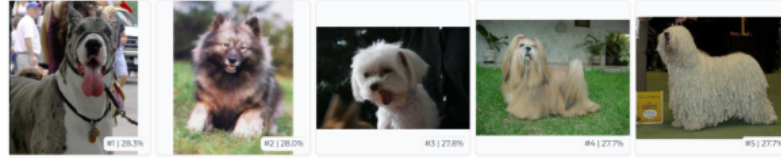
Amanuel Gizachew Abebe
amanuel.g.abebel@gmail.com
Shaggar Institute of Technology (SIT)
Shaggar City, Ethiopia

Simret A Gebreegziabher
sgebreeg@nd.edu
University of Notre Dame
Notre Dame, USA

Subjective Query: an aggressive dog

$P=\{\text{angry, fierce, showing teeth, growling...}\}$ $N=\{\text{friendly, calm, relaxed...}\}$

Baseline



Feedback

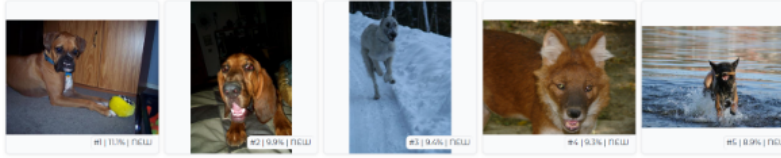


Figure 1: Qualitative comparison for the query “an aggressive looking dog” (baseline CLIP vs. steered retrieval using per-concept weighting).

Abstract

Vision–language models such as CLIP enable open-vocabulary image retrieval, but they are optimized for objective, widely agreed-upon concepts and often fail on subjective or user-specific notions such as “looking guilty.” For these interpretive concepts, there may be no single ground truth: intent varies across users and contexts, making offline fine-tuning expensive, brittle, and potentially normatively restrictive. Motivated by pluralistic alignment, we study an interaction-time alternative that keeps CLIP frozen and lets users negotiate meaning through natural-language critiques. Starting from an initial query embedding q , we parse user feedback into positive and negative attributes and steer the query in CLIP’s text embedding space via $q' = q + \alpha \text{embed}(\text{positive}) - \beta \text{embed}(\text{negative})$. We compare this steering rule against five other inference-time methods on curated queries ($N=22$) on three datasets. Across a mix of objective and subjective queries, we find that mean Precision@5 improves, with the largest gains being on subjective concepts as opposed to objective ones. These results suggest that lightweight, interpretable steering can function as a practical interaction design

*Both authors contributed equally to this research.

for bi-directional human-AI alignment, shifting alignment from a training-time objective to an interactive process.

Keywords

Human-AI Alignment, Interactive Retrieval, Vision-Language Models, Subjective Concepts, Human Agency

ACM Reference Format:

Mihret Agegnehu Bekele, Dawit Getahun Mengistu, Kidus Paulos, Natnael Abayneh, Amanuel Gizachew Abebe, and Simret A Gebreegziabher. 2026. Steering Vision Models towards Subjective Concepts with Language. In *BiAlign '26: CHI '26 Workshop on Human-AI Interaction Alignment*, April 13, 2026, Barcelona, Spain. ACM, New York, NY, USA, 11 pages.

1 Introduction

Vision–language models such as CLIP [10] increasingly mediate what people see from search and curation to moderation and decision support. Yet many queries may not always be objective facts (“a red car”) but interpretations (“looks guilty,” “seems friendly,” “aggressive-looking”). These concepts are inherently subjective, context-dependent, and often contested. Different users can legitimately disagree, and even a single user may revise their intent after

seeing results. This creates a core alignment problem: how can a system remain useful and accountable when the target concept is not fixed?

Common responses to misalignment implicitly assume that desired behavior can be specified offline and then “locked in.” For subjective retrieval, this is a poor fit. Baking a single normative definition into model parameters can suppress nuance and erase pluralism. Pluralistic alignment argues that interactive systems should support multiple legitimate objectives rather than converging to a single one [1]. For vision–language retrieval, this means that alignment should be treated as an interaction design problem. Specifically, the system must help users express what they mean in the moment and update their behavior accordingly.

Figure 2 illustrates how our approach can serve different user definitions of the same concept at inference time without retraining. Two users can issue the same natural-language query (“an exciting action scene”) but supply different critique profiles that operationalize different meanings of *exciting*. One user may equate excitement with speed and motion (e.g., “more running/jumping,” “less stillness”), while another equates it with danger and intensity (e.g., “more crash/fire,” “less safe/quiet”). Applying the additive update separately for each profile yields different steered query embeddings, producing distinct top- k result sets from the same frozen CLIP backbone. In this way, pluralism becomes a lightweight, interaction-time definition rather than a single training-time ground truth.

To address this, we explore different inference-time approaches that keep the base model frozen while adapting retrieval behavior through structured interaction. The user starts with an initial query and then provides natural-language critiques (e.g., “more tense posture,” “less smiling”), which we parse into positive and negative attributes. Given an initial query embedding q , we steer the embedding as

$$q' = q + \alpha \text{ embed(positive)} - \beta \text{ embed(negative)},$$

enabling multi-round refinement without retraining. Because the update is additive and interpretable, it offers a concrete interaction useful for bi-directional human–AI alignment. Users adjust the model’s behavior, while the model’s responses help them clarify and refine their intent.

We evaluate this additive feedback rule alongside other approaches including contrastive subspace steering, energy-based steering, per-concept weighting, and SAE-PRF steering on curated subsets of Flickr8k [4], CelebA [9], and Stanford Dogs [7] datasets. Beyond retrieval gains, our results highlight a broader implication and opportunities for interaction design. When concepts are subjective, alignment can emerge through interaction. Lightweight, interpretable steering mechanisms could potentially support co-adaptive feedback loops that preserve user agency while keeping the underlying model stable, making space for pluralistic, user-specific notions of relevance.

2 Related Work

We situate our work across three strands: vision–language alignment, interactive information retrieval, and inference-time representation steering.

2.1 Vision-Language Alignment

Current state-of-the-art vision-language models, such as CLIP [10], align images and text into a shared embedding space using contrastive learning. While effective for zero-shot classification, these models are static once trained. They represent a snapshot of the data they were trained on. Previous efforts to improve alignment have focused on adapter layers [5] or learned prompt/context tuning [17, 18] (i.e., optimizing continuous prompt vectors for CLIP-like models, as in CoOp/CoCoOp), as well as soft prompt tuning for frozen language models [8]. However, these approaches still require optimization on task-specific data (often labeled) and are therefore ill-suited for on-the-fly, user-specific subjective concepts (e.g., “vibe” or “guilty look”), where training data is scarce, and user intent may shift rapidly.

2.2 Interactive Information Retrieval

The field of HCI has long studied relevance feedback in search engines. Traditional systems use click-through data (User clicked Image A, ignored Image B) to re-rank results. While effective, binary clicks are a low-bandwidth signal. More recently, Agile Modeling [14] has suggested that users should be able to define new concepts on the fly. We therefore examine natural language as the primary steering mechanism, making the feedback loop more expressive than simple clicks. By allowing users to say why an image is wrong (e.g., “too dark”), we capture semantic directionality that clicks cannot convey.

2.3 Steering and Test-Time Adaptation in Representation Space

Recent works have explored *representation steering*—controlling model behavior at inference time by intervening on internal activations or learned features, without updating the base model weights. For vision–language models, Joseph et al. [6] study steerability in CLIP’s vision transformer using sparse autoencoders (SAEs) and quantify which features can be manipulated to induce targeted output changes. Separately, Zaigrajew et al. [16] introduce hierarchical SAEs for CLIP and release pretrained SAE weights, enabling downstream work to operate in SAE space without training an SAE from scratch. Complementarily, SteerVLM [13] proposes a lightweight module for activation steering in VLMs that modulates cross-modal activations to improve instruction adherence while preserving general performance. In our setting, we use pretrained hierarchical SAE weights [16] as a latent coordinate system for CLIP embeddings, and apply a simple pseudo-relevance feedback (PRF) update in SAE residual space—using the baseline top- K retrieved images as pseudo-positives and steering the query latent toward their shared direction—then decode back to CLIP space for a second retrieval pass.

3 Inference-Time Steering

3.1 Refined Query via Natural-Language Feedback

To allow lightweight steering, we refine the query embedding by shifting it toward positive attributes and away from negative ones via vector arithmetic in CLIP’s text embedding space.



Figure 2: Serving different user definitions of the same concept at inference time. Starting from the same baseline CLIP query for “an exciting action scene” (top row), two users provide different critique profiles (speed/motion vs. danger/intensity). The same frozen model can then be steered to return different results for each user (middle and bottom rows), reflecting user-specific intent without retraining.

We formalize the update as follows:

$$V'_q = \text{normalize} \left(V_q + \alpha \sum_{p \in P} \text{Enc}(p) - \beta \sum_{n \in N} \text{Enc}(n) \right) \quad (1)$$

where V_q is the initial query embedding, P and N are sets of positive and negative attribute phrases, $\text{Enc}(\cdot)$ is the frozen CLIP text encoder, and α, β control the update strength. This additive update is closely related to prompt engineering and prompt ensembling approaches for CLIP [17, 18].

3.2 Per-Concept Attribute Weighting

Subjective concepts are rarely defined by attributes of equal importance [11]. To account for this granularity, we extend the linear additive feedback rule to assign each attribute an independent weight. Specifically, we introduce per-attribute coefficients w_i^+ and w_j^- for positive and negative attributes, respectively, and redefine the energy as:

$$V'_q = \text{normalize} \left(V_q + \sum_i w_i^+ \text{Enc}(p_i) - \sum_j w_j^- \text{Enc}(n_j) \right), \quad (2)$$

These weights may be initialized using heuristic signals such as similarity between the attribute and the original query or confidence scores derived from language-based analysis. Crucially, all weights are interpretable and can be directly adjusted by the user, enabling fine-grained control over how different semantic factors contribute to the steering process. Per-concept weighting subsumes the global scalar parameters α and β , while offering greater flexibility for modeling nuanced subjective intent.

3.3 Contrastive Concept Subspace Steering

While the additive steering rule in Equation (1) provides an effective first-order mechanism for incorporating user feedback, subjective

concepts often span multiple semantic dimensions (e.g., posture, expression, and context). To better capture this structure, we introduce a low-dimensional *contrastive concept subspace* that separates positive and negative attributes in a more expressive yet interpretable manner.

Let $\{\text{Enc}(p_i)\}$ and $\{\text{Enc}(n_j)\}$ denote the CLIP text embeddings of positive and negative attribute phrases, respectively. We learn a linear projection matrix $W \in \mathbb{R}^{d \times k}$, with $k \ll d$, that maximally separates these two sets under a contrastive objective similar to Fisher discriminant analysis [3]. This projection defines a concept-specific coordinate system in which each axis corresponds to a distinct semantic factor underlying the subjective concept.

Given a query embedding q , we project it into the concept subspace as:

$$z_0 = W^\top q. \quad (3)$$

We then compute centroid-based steering:

$$\hat{d} = \text{normalize} \left(\frac{1}{|P|} \sum_{p_i \in P} W^\top \text{Enc}(p_i) - \frac{1}{|N|} \sum_{n_j \in N} W^\top \text{Enc}(n_j) \right), \quad (4)$$

$$z' = z_0 + \alpha \hat{d}. \quad (5)$$

and map back for retrieval as $q' = \text{normalize}(Wz')$.

3.4 Energy-Based Steering in the Concept Subspace

Rather than applying a fixed linear update in the concept subspace, we formulate steering as the minimization of an energy function that explicitly encodes the user’s intent, following standard energy-based modeling formulations [2]. This allows smooth and controlled adaptation while preserving the original semantics of the query.

Let $z_{p_i} = W^\top \text{Enc}(p_i)$ and $z_{n_j} = W^\top \text{Enc}(n_j)$ denote the projected attribute embeddings. We define the following energy over subspace

representations z :

$$E(z) = -\alpha \sum_i \text{sim}(z, z_{p_i}) + \beta \sum_j \text{sim}(z, z_{n_j}) + \lambda \|z - z_0\|^2, \quad (6)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and λ controls deviation from the original query.

This formulation directly operationalizes user intent. Positive attributes contribute negative energy, creating attractive regions in the subspace that pull the query toward desired semantic directions. Negative attributes contribute positive energy, forming repulsive regions that push the query away from undesired characteristics. The regularization term $\lambda \|z - z_0\|^2$ anchors the update to the original query representation, ensuring that refinement reflects modification rather than semantic drift.

Minimizing $E(z)$ therefore corresponds to finding a representation that best satisfies the user’s articulated preferences while remaining grounded in their initial request. We perform a small number of gradient steps to obtain z^* , enabling incremental, interpretable refinement rather than abrupt shifts.

The steered query embedding is then reconstructed in the original CLIP space as:

$$q' = q + W(z^* - z_0). \quad (7)$$

This energy-based formulation generalizes the additive update in Equation (1), providing a principled mechanism for negotiating multiple, potentially competing subjective attributes in a controlled manner.

3.5 SAE-PRF Steering (Pseudo-Relevance Feedback in SAE Space)

Lastly, we consider an SAE-based pseudo-relevance feedback (PRF) method that steers a query based on the baseline top- K retrieved images, treating them as pseudo-positives. This approach gives better control over the model’s internal activations. Let $q \in \mathbb{R}^d$ be a CLIP query embedding, and $x_i \in \mathbb{R}^d$ be CLIP image embeddings. Let an SAE encoder/decoder map between CLIP space and SAE latent space: $z = E(q) \in \mathbb{R}^D$, $\hat{q} = D(z) \in \mathbb{R}^d$. We use a dataset-level mean latent activation $\mu \in \mathbb{R}^D$ and define residual latents $r = z - \mu$.

Algorithm 1 shows the algorithm for SAE-space PRF. Let I_K be indices of the baseline top- K images. $\bar{r}$ summarizes what the baseline top- K results have in common in SAE residual space, and steering moves the query toward that shared latent signature.

4 Experimental Setup

We implemented our pipeline using OpenAI’s CLIP (ViT-B/32) model and PyTorch. All images were preprocessed to a resolution of 224×224 pixels and normalized according to CLIP’s training distribution. We compared the methods discussed in Section §3 against a vanilla CLIP baseline. We use three datasets for this experiment, focusing on a subset of 22 queries spanning subjective and objective queries. We report performance measured by Precision@K with averaged human annotation to account for the subjective nature of the queries. All steering approaches are computationally lightweight and did not require specialized hardware. Experiments were run on Google Colab¹ and a Intel(R) Core(TM) i7-13620H CPU, 16

¹<https://colab.research.google.com/>

Algorithm 1 Pseudo-Relevance Feedback with Residual Update

```

1: Input: Query  $q$ , Corpus  $\{x_i\}$ , Parameters  $K, M, \alpha$ 
2: Output: Retrieved documents based on updated query  $q'$ 
   ▷ 1. Baseline Retrieval
3:  $I_K \leftarrow \text{TopK}(\cos(q, x_i))$ 
   ▷ 2. Encode and Mean-Center
4:  $z_q \leftarrow E(q)$ ,  $r_q \leftarrow z_q - \mu$ 
5: for  $i \in I_K$  do
6:    $z_i \leftarrow E(x_i)$ 
7:    $r_i \leftarrow z_i - \mu$ 
8: end for
   ▷ 3. PRF Direction
9:  $\bar{r} \leftarrow \frac{1}{K} \sum_{i \in I_K} r_i$ 
   ▷ 4. Feature Selection
10:  $S \leftarrow \text{TopMIndices}(|\bar{r}|)$ 
   ▷ 5. Residual Update
11: for  $j \in \{1, \dots, \text{dim}\}$  do
12:   if  $j \in S$  then
13:      $(r'_q)_j \leftarrow (r_q)_j + \alpha \cdot \bar{r}_j$ 
14:   else
15:      $(r'_q)_j \leftarrow (r_q)_j$ 
16:   end if
17: end for
   ▷ 6. Decode and Retrieve
18:  $z'_q \leftarrow r'_q + \mu$ 
19:  $q' \leftarrow D(z'_q)$ 
20: return Retrieve( $q'$ )

```

GB of system RAM, and integrated Intel(R) UHD Graphics utilizing 7.8 GB of shared GPU memory.

We have released a web-based interface that allows for real-time interaction. We include a public demo for readers to explore the steering loop and qualitative results of the methods.²

4.1 Datasets

We curated subsets from three public datasets to ensure manageable inference times for interactive experiments. We selected the following datasets specifically for their high potential for subjective interpretation.

- **Flickr8k** [4]: A subset of 300 images depicting diverse scenes, used for ambiguous concepts.
- **CelebA** [9]: A subset of 500 face images, chosen to test subjective facial attribute queries (e.g., “suspicious”).
- **Stanford Dogs** [7]: A subset of 500 images covering 120 breeds, used for fine-grained behavioral traits.

4.2 Protocol

We evaluated 22 queries with a mix of objective (e.g., “wearing glasses”) and subjective (e.g., “looking guilty”) queries. See Table 1 for detailed queries. For each query, we measured Precision@5 (P@5) and Precision@10 (P@10) using the CLIP model. We then

²Demo: https://huggingface.co/spaces/mihretgold/Steering_Vision_Language_Models_with_Subjective_Concepts

Table 1: Per-Query Precision@5 Comparison Across Methods

Query	Baseline CLIP	Linear Steering	Contrastive Subspace	Energy-Based	Per-Concept Weighted	SAE PRF Steering
<i>Stanford Dogs</i>						
A golden retriever (<i>Objective</i>)	0.80	0.80	0.80	0.40	0.60	0.40
Dog on the beach (<i>Objective</i>)	1.00	1.00	0.60	0.80	0.60	1.00
Dog looking guilty (<i>Subjective</i>)	0.80	1.00	0.80	0.80	0.80	1.00
Friendly looking dog (<i>Subjective</i>)	0.80	0.40	0.60	0.80	0.80	0.60
Aggressive looking dog (<i>Subjective</i>)	0.60	0.80	0.80	1.00	1.00	0.80
Nervous looking dog (<i>Subjective</i>)	0.60	0.60	0.80	0.80	0.80	0.60
Hyper active dog (<i>Subjective</i>)	0.60	0.60	1.00	1.00	1.00	0.60
<i>Flickr8k</i>						
A person riding a bicycle (<i>Objective</i>)	0.60	0.60	0.60	0.60	0.60	0.60
A dog playing (<i>Objective</i>)	0.80	1.00	1.00	0.80	1.00	0.80
An exciting action scene (<i>Subjective</i>)	0.80	1.00	1.00	0.80	0.80	1.00
A joyful moment (<i>Subjective</i>)	0.80	0.80	0.80	0.60	0.80	0.80
A kid having fun (<i>Subjective</i>)	0.80	1.00	0.80	0.80	0.80	0.80
Peaceful scene (<i>Subjective</i>)	1.00	1.00	0.80	1.00	1.00	1.00
A photo with motion (<i>Subjective</i>)	0.60	0.80	0.80	0.80	0.60	0.60
<i>CelebA</i>						
Wearing eyeglasses (<i>Objective</i>)	1.00	1.00	1.00	1.00	1.00	1.00
A person smiling (<i>Objective</i>)	1.00	1.00	1.00	1.00	1.00	1.00
Looking guilty (<i>Subjective</i>)	0.40	0.60	0.80	0.40	0.80	0.00
Looking happy (<i>Subjective</i>)	1.00	1.00	1.00	1.00	1.00	1.00
Looking sad (<i>Subjective</i>)	0.60	0.80	0.40	0.60	0.80	0.60
Looking suspicious (<i>Subjective</i>)	0.60	1.00	0.40	0.60	0.40	0.60
Looking tired (<i>Subjective</i>)	0.60	0.80	0.60	0.60	0.60	0.20
Looking confident (<i>Subjective</i>)	0.80	1.00	0.80	1.00	1.00	1.00

applied natural language feedback and measured the metrics again on the steered results.

To obtain relevance labels, six annotators participated, rotating across datasets. For each evaluation condition, annotators were shown the exact retrieval output produced from the same textual query and the same positive/negative descriptor set used to generate that output. They then judged each returned image as relevant or non-relevant.

Each image in a given condition was labeled independently by three annotators (1 = relevant, 0 = not relevant), with no communication during labeling. We used majority vote only to produce a stable aggregate label for metric computation at the image level under that fixed condition (query + descriptor profile + method), while preserving individual-level ratings for agreement analysis.

5 Results

Our experiments show that natural-language feedback improves retrieval performance primarily for subjective queries, where intent is underspecified by a single query. Figure 1 provides a qualitative example for the query “an aggressive looking dog,” illustrating how per-concept weighting re-ranks results toward images that better match a user’s interpretation as expressed through positive and negative attributes.

5.1 Annotation Agreement and Overall Performance

Annotator agreement varied substantially by query type. Annotators achieved excellent consensus with a Kappa coefficient of 0.824 (91.7% agreement) on objective queries, while showed only moderate agreement (0.506 kappa, 68.6% agreement) on subjective queries.

This highlights that subjective queries tap into personal interpretations. Table 2 reports aggregate results across both subjective and objective queries. Steering methods consistently outperform the baseline, with the strongest gains coming from Linear and Per-concept Weighted steering. The moderate improvements reflect the difficulty of grounding concepts like “guilty” or “nervous”, where it is non-trivial to capture human interpretation. We find statistically significant improvements for the two techniques, while we do not find statistical significance for Contrastive, Energy-Based, and PRF methods.

5.2 Performance by Query Type

CLIP’s baseline retrieval performance on objective queries was already strong ($P@5 = 0.867$), and no steering method improves it significantly. On subjective queries however, the baseline drops to $P@5 = 0.712$, and the steering techniques show performance gains, with the largest absolute gains on queries like “hyperactive dog” and “aggressive dog” (+0.4 $P@5$). This confirms that steering primarily benefits concepts poorly captured by CLIP’s pretrained representations. All significance observed in the aggregate results is driven entirely by subjective queries.

Across all queries, Linear Steering significantly improves both Precision@5 ($t = 2.664$, $p = 0.0145$) and Precision@10 ($t = 3.578$, $p = 0.0018$), increasing mean $P@5$ from 0.755 to 0.845. Per-Concept Weighting yields a significant gain for $P@10$ ($t = 2.630$, $p = 0.0156$) but not for $P@5$, suggesting that weighting may be especially useful for broadening the top of the ranked list rather than consistently improving the very top results. The remaining methods (Contrastive Subspace, Energy-Based Subspace, and SAE-PRF) show small, non-significant changes ($p > 0.05$). Notably, SAE-PRF seemed to slightly degrade performance relative to baseline ($P@5 = 0.727$ vs. 0.755).

Table 2: Aggregated retrieval performance (mean $\pm$ std).

Method	Type	P@5	P@10	MRR@5	MRR@10
Baseline	Obj	0.867 (0.163)	0.800 (0.200)	0.917 (0.204)	0.917 (0.204)
	Subj	0.712 (0.163)	0.625 (0.279)	0.849 (0.276)	0.849 (0.276)
Contrastive	Obj	0.833 (0.197)	0.817 (0.160)	0.917 (0.204)	0.917 (0.204)
	Subj	0.762 (0.182)	0.706 (0.221)	0.844 (0.239)	0.844 (0.239)
Linear	Obj	0.900 (0.167)	0.833 (0.207)	0.917 (0.204)	0.917 (0.204)
	Subj	0.825 (0.191)	0.738 (0.236)	0.969 (0.125)	0.969 (0.125)
Energy	Obj	0.767 (0.234)	0.767 (0.186)	0.917 (0.204)	0.917 (0.204)
	Subj	0.788 (0.186)	0.738 (0.206)	0.969 (0.125)	0.969 (0.125)
Weighting	Obj	0.800 (0.219)	0.850 (0.176)	0.917 (0.204)	0.917 (0.204)
	Subj	0.812 (0.171)	0.769 (0.233)	0.938 (0.171)	0.938 (0.171)
PRF	Obj	0.800 (0.253)	0.717 (0.279)	0.917 (0.204)	0.917 (0.204)
	Subj	0.700 (0.292)	0.656 (0.299)	0.865 (0.306)	0.875 (0.276)

Table 3: Significance tests vs. Baseline CLIP (all queries): paired t -test and Wilcoxon signed-rank test. Asterisks indicate $p < 0.05$.

Method	P@5				P@10			
	t	p	W	p	t	p	W	p
Contrastive	1.000	0.333	14.0	0.163	1.282	0.219	11.0	0.172
Linear	2.522	0.024 *	10.5	0.042 *	4.137	0.001 *	3.0	0.004 *
Energy	1.861	0.083	3.0	0.111	1.820	0.089	21.0	0.086
Weighting	2.236	0.041 *	1.5	0.033 *	2.556	0.022 *	6.0	0.016 *
PRF	-0.269	0.792	10.5	0.551	1.098	0.290	16.0	0.230

Method	MRR@5				MRR@10			
	t	p	W	p	t	p	W	p
Contrastive	-0.064	0.950	10.5	1.000	-0.064	0.950	10.5	1.000
Linear	1.502	0.154	2.0	0.131	1.502	0.154	2.0	0.131
Energy	1.502	0.154	2.0	0.131	1.502	0.154	2.0	0.131
Weighting	1.000	0.333	5.0	0.236	1.000	0.333	5.0	0.236
PRF	0.221	0.828	4.5	0.854	0.379	0.710	4.5	0.854

5.3 User Study

The evaluation we have seen so far relied on six researcher-annotators with majority voting, providing a controlled but narrow perspective. To validate whether Linear Steering improves retrieval quality from the standpoint of a broader user population, we conducted a follow-up study with 15 external participants who independently judged the relevance of retrieved images.

5.3.1 Study Design. Each participant evaluated the top-5 retrieved images for all 22 queries under two conditions: *Baseline CLIP* (vanilla cosine-similarity retrieval) and *LLM Linear Steering* (Equation 1). For Linear Steering, an LLM proposed initial positive and negative attribute descriptors for each participant. Participants iteratively edited this set by removing irrelevant attributes and adding or rephrasing attributes to better reflect their intent, using retrieval updates as feedback. Participants then provided binary relevance judgments (relevant/not relevant) for each image, from which we compute per-annotator Precision@5 and MRR@5 (Mean Reciprocal Rank computed after 5 images). Additional details in Appendix A.

A key distinction from the earlier evaluation is that *Baseline CLIP* returns the same five images to every participant for a given query, whereas *Linear Steering* produces personalized results because each participant’s critique profile yields a different steered query

Table 4: User study results ($N = 15$ participants, 22 queries). P@5 improvement is significant; MRR@5 improvement is not.

Method	Type	P@5	MRR@5
Baseline CLIP	Obj	0.820 (0.098)	0.967 (0.024)
	Subj	0.729 (0.174)	0.859 (0.168)
Linear Steering	Obj	0.883 (0.097)	0.956 (0.031)
	Subj	0.824 (0.121)	0.906 (0.098)
Δ (Linear – CLIP)	Obj	+0.063 [*]	−0.011
	Subj	+0.095 ^{**}	+0.047

All queries: $\Delta P@5 = +0.090$, $t = 4.20$, $p < 0.001$, Cohen’s $d = 0.89$
Win/Tie/Loss (P@5): 19 / 2 / 1 ^{*} $p < 0.05$ ^{**} $p < 0.01$

Table 5: Image-aware inter-annotator agreement. Linear Steering’s low image overlap on subjective queries reflects personalization; naïve κ is misleading when annotators evaluate different images.

Method	Type	Overlap	Same-Img κ	Interp.
Baseline CLIP	Obj	100%	0.486	Moderate
	Subj	100%	0.321	Fair
Linear Steering	Obj	40%	— [†]	—
	Subj	23%	— [†]	—

[†] Too few same-image pairs for reliable κ estimation.

Same-image disagreement rates are comparable across methods (see §5.3.3).

embedding. This asymmetry is central to interpreting both precision and inter-annotator agreement.

5.3.2 Study Results. Table 4 reports aggregate performance. Linear Steering improves mean P@5 from 0.745 to 0.835 (+0.090, 12.1% relative improvement) with a large effect size (Cohen’s $d = 0.89$). The improvement is highly significant under both a paired t -test ($t = 4.20$, $p < 0.001$) and a Wilcoxon signed-rank test ($W = 10.5$, $p < 0.001$). MRR@5 also increases (+0.036), though the gain is not statistically significant ($p = 0.12$), likely because the baseline already achieves high MRR (0.879), leaving limited room for improvement at rank 1.

Consistent with our earlier findings, the improvement is larger for subjective queries ($\Delta P@5 = +0.095$, $p = 0.002$) than for objective queries (+0.063, $p = 0.046$), confirming that steering primarily benefits concepts where CLIP’s text–image alignment is weakest. Every one of the 15 participants individually achieves equal or higher P@5 under Linear Steering ($\Delta P@5$ range: 0.00 to +0.18), demonstrating that the improvement is robust across annotators with different levels of strictness.

User Experience Ratings. Participants rated four experience dimensions at the end of the study Figure 3 on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree; neutral midpoint = 4): *System Understood* (“The system understood what I meant”), *Felt in Control* (“I felt in control of the results”), *Satisfied w/ Results* (“I am satisfied with the results”), and *Interaction Frustrating* (“The interaction was frustrating”). For the first three dimensions, higher scores indicate better perceived alignment, agency, and satisfaction; for

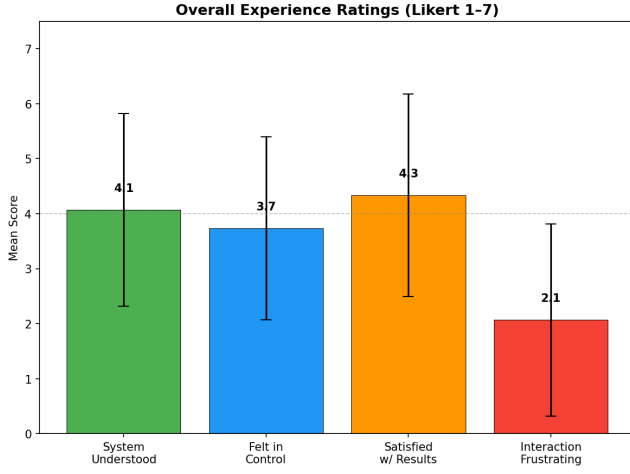


Figure 3: User experience ratings (N=15). Mean participant ratings across four experience dimensions on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). Error bars indicate standard deviation; the dashed line marks the neutral midpoint (4).

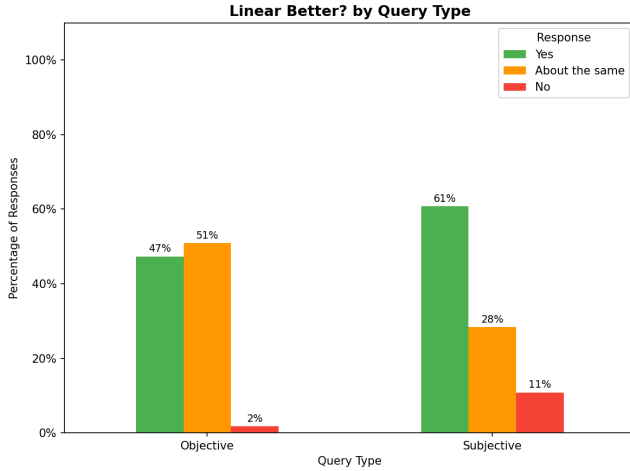


Figure 4: Objective vs. subjective query comparison (N=15). Percentage of participant responses indicating whether Linear Steering produced better top-5 results than the Baseline, grouped by query type (Objective vs. Subjective).

frustration, lower scores are better. Error bars denote the standard deviation across participants.

Objective vs. Subjective Query Analysis. Figure 4 summarizes per-query comparison responses to “Did Linear Steering give better top-5 results than Baseline?” grouped by objective (visually verifiable) and subjective (emotional/abstract) queries. A higher proportion of “Linear better” responses for subjective queries supports our central hypothesis that attribute-based steering is most beneficial for abstract concepts where zero-shot CLIP retrieval is less reliable,

while objective queries are more likely to be rated “about the same” when the baseline already performs well.

5.3.3 Image-Aware Inter-Annotator Agreement. Comparing inter-annotator agreement across the two methods requires care, because participants are not always evaluating the same images. Table 5 reports image overlap rates and agreement statistics. For Baseline CLIP, every annotator pair shares 100% of the retrieved images, and Cohen’s κ can be computed directly: $\kappa = 0.486$ (moderate) on objective queries and $\kappa = 0.321$ (fair) on subjective queries.

For Linear Steering, image overlap drops substantially—to ~40% on objective queries and ~23% on subjective queries—because each participant’s personalized critique profile produces a different set of retrieved images. This low overlap makes naïve κ (computed over all annotator pairs, including those who saw entirely different images) misleading, as it conflates genuine evaluative disagreement with stimulus variation. When restricted to same-image pairs, the per-query disagreement rates for Linear Steering are comparable to those of Baseline CLIP (e.g., “looking suspicious”: 49.1% vs. 48.9%; “aggressive looking dog”: 41.5% vs. 36.1%), indicating that when annotators *do* evaluate the same image, they agree at similar rates regardless of retrieval method.

Crucially, the low image overlap for Linear Steering on subjective queries is not a limitation but *evidence that personalization is functioning as intended*: different users receive different results tailored to their individual steering profiles, while per-annotator precision remains high (mean P@5 = 0.824 on subjective queries). This is consistent with the pluralistic alignment perspective motivating our work (§1): rather than converging on a single “correct” result set, the system supports multiple legitimate interpretations of subjective concepts.

6 Discussion

We compared six inference-time steering strategies across 22 queries and three datasets. The results reveal that not all methods are equally effective. Our findings demonstrate that we can effectively allow users to steer vision language models through inference-time steering. We find that a well-tuned additive update with user defined defining attributes is both statistically robust and practical. Per-concept weighting offers additional control for users who wish to prioritize specific attributes, particularly for broadening relevant results.

6.1 Towards Bi-directional Model Steering

Our experiments show that user-defined concepts can steer a model in a single-turn format where the user expresses intent through adjectives and the system returns a re-ranked, “steered” output. We view this interaction as a starting point for *bi-directional* alignment, where (i) people shape the model’s behavior through feedback and (ii) the model helps people articulate, test, and refine their own intent. In other words, alignment becomes a joint process of meaning-making rather than a one-sided optimization performed only at training time.

According to Norman’s theory of the Gulf of Execution and Evaluation [15], a primary challenge in HCI and human–AI collaboration is the distance between a user’s goals and the system’s

observable state. In standard CLIP-style retrieval, the *Gulf of Execution* for subjective concepts (e.g., “looking guilty”) is wide because users cannot easily translate a felt, context-dependent impression into the model’s fixed embedding space. The inference-time steering mechanisms narrow this gulf by providing an interaction primitive for direct manipulation of the latent space. Users can add positive and negative descriptors, see the effect on the ranked results, and iterate. In this sense, the user is not just searching but actively shaping the decision boundary of relevance.

However, true bi-directional steering requires more than accepting user-provided attributes. It also requires the system to scaffold the interaction by making the model’s internal semantics legible and by eliciting feedback in forms that are easy for users to provide. We outline three concrete design opportunities for closing both gulfs:

- **Closing the Gulf of Execution via critique scaffolding.** Users may know what they want but struggle to describe it. The system can propose candidate attributes (e.g., “more formal,” “less smiling,” “darker lighting”) extracted from differences between relevant and non-relevant results, letting users accept, reject, or rephrase them. This turns free-form critique into a mixed-initiative interaction.
- **Closing the Gulf of Evaluation via semantic explanations.** After applying an update, the system can summarize what changed in the query embedding (e.g., top attributes increased/decreased) and provide counterfactual previews (“if we down-weight *smiling*, these items drop”). Such feedback helps users build a mental model of how language maps to retrieval outcomes.
- **Supporting multi-turn, memory-aware steering.** Subjective concepts often stabilize over a session but vary across users. A bi-directional system can maintain a short-term “session concept” that accumulates confirmed attributes, while still allowing rapid resets and alternatives (“two interpretations of *guilty*: 1. nervous posture, 2. avoiding eye contact”). This supports exploration without forcing premature commitment.

6.2 Human Agency and the Locus of Control

Pluralistic alignment states that in many interactive settings there is no single correct objective, and systems should support multiple legitimate interpretations of what counts as relevant or appropriate [1]. Inference-time steering advances this goal by making the alignment target explicit and editable at interaction time. Rather than being passive recipients of the model’s pre-trained biases, users act as co-designers of the retrieval criteria in real time. They can specify what should matter for the current task, and revise those preferences when the system misinterprets them.

This perspective is especially important for subjective concepts, where there may be no single ground truth. Two users can reasonably disagree about what counts as “professional,” “friendly,” or “guilty,” and even the same user can shift criteria across contexts. Steering supports this plurality by allowing multiple, local alignments to coexist. The same underlying model can serve different values without requiring global fine-tuning. This aligns with value-centered alignment perspectives [12] by treating values as

interaction-time commitments rather than static developer choices, and it complements pluralistic alignment by making that plurality operational through inference-time definitions.

We acknowledge this increased user control introduces new interaction design risks and responsibilities. First, *choice overload* can occur if users are asked to provide too many attributes without guidance. Second, *illusion of control* is possible if the interface suggests agency while the model cannot faithfully represent certain attributes. Transparency about failure modes and limitations is therefore critical.

7 Limitations and Future Work

Our study has four primary limitations. First, statistical power is constrained by the scale of our evaluation, which used only 22 queries across subsets of Flickr8k, CelebA, and Stanford Dogs. That said, the large observed effect size (Cohen’s $d = 0.89$) suggests the primary finding is robust even at this scale. Future work should scale to 100+ queries spanning a wider variety of subjective categories. Second, while we used majority voting to aggregate three ratings per image under each fixed condition (query + descriptor profile + method), this aggregation can smooth out individual variance. Thus, condition-level aggregate labels remain an imperfect proxy for user-specific intent, especially for subjective concepts (consistent with our $\kappa = 0.506$). Third, our follow-up user study (15 participants) evaluates perceived relevance but it does not capture longer-term concept drift, learning effects, or how users iteratively refine critiques over multiple rounds in realistic tasks. Fourth, the SAE-PRF method degraded performance relative to the baseline, suggesting that unsupervised feedback in pretrained SAE latent spaces may be unreliable when the initial retrieval set is heterogeneous.

Future work will extend this evaluation by recruiting a larger and more diverse participant sample, measuring user satisfaction and perceived control in addition to relevance judgments, and studying multi-round refinement behavior in realistic tasks using our web-based interface. Additionally, we aim to investigate why lightweight linear steering outperformed complex subspace projections, to develop hybrid methods that combine the interpretability of linear updates with the semantic disentanglement of sparse autoencoders.

8 Conclusion

We conducted a comparative evaluation of inference-time steering strategies for aligning vision–language retrieval with subjective concepts using natural-language feedback, while keeping CLIP frozen. Across three datasets and 22 queries, our results show that simple additive Linear Steering yields the most consistent and statistically significant improvements—particularly for Precision@5 and Precision@10. In contrast, more complex subspace variants and SAE-PRF pseudo-relevance feedback methods did not reliably outperform this lightweight approach.

Overall, our findings demonstrate that interpretable, language-based query steering is a practical and effective mechanism for adapting retrieval systems to subjective, context-dependent user intent.

References

- [1] Daiwei Chen, Yi Chen, Aniket Rege, and Ramya Korlakai Vinayak. 2024. Pal: Pluralistic alignment framework for learning from heterogeneous preferences. *arXiv preprint arXiv:2406.08469* (2024).
- [2] Stanford CS236. 2024. CS236 Lecture 12: Energy-Based Models (slides). https://deepgenerativemodels.github.io/assets/slides/cs236_lecture12.pdf. Accessed 2026-02-10.
- [3] José Guillermo González Mora, Hiram Ponce, and Lourdes Martínez-Villaseñor. 2025. Contrastive Steering Vectors for Autoencoder Explainability. *Electronics* 14, 18 (2025), 3586.
- [4] Micah Hodosh, Peter Young, and Julia Hockenmaier. 2013. Framing Image Description as a Ranking Task: Data, Models and Evaluation Metrics. *Journal of Artificial Intelligence Research* (2013).
- [5] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for NLP. In *International conference on machine learning*. PMLR, 2790–2799.
- [6] Sonia Joseph, Praneet Suresh, Ethan Goldfarb, Lorenz Hufe, Yossi Gandelsman, Robert Graham, Danilo Bzdok, Wojciech Samek, and Blake Aaron Richards. 2025. Steering CLIP’s vision transformer with sparse autoencoders. *arXiv preprint arXiv:2504.08729* (2025).
- [7] Aditya Khosla, Nityananda Jayadevaprakash, Bangpeng Yao, and Li Fei-Fei. 2011. Novel dataset for fine-grained image categorization. In *First Workshop on Fine-Grained Visual Categorization, IEEE Conference on Computer Vision and Pattern Recognition*.
- [8] Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the 2021 conference on empirical methods in natural language processing*. 3045–3059.
- [9] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. 2015. Deep learning face attributes in the wild. In *Proceedings of the IEEE international conference on computer vision*. 3730–3738.
- [10] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pami Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*. PMLR, 8748–8763.
- [11] Eleanor Rosch and Carolyn B Mervis. 1975. Family resemblances: Studies in the internal structure of categories. *Cognitive psychology* 7, 4 (1975), 573–605.
- [12] Hua Shen, Tiffany Kneare, Reshmi Ghosh, Yu-Ju Yang, Nicholas Clark, Tanu Mitra, and Yun Huang. 2025. ValueCompass: A framework for measuring contextual value alignment between human and LLMs. In *Proceedings of the 9th Widening NLP Workshop*. 75–86.
- [13] Anushka Sivakumar, Andrew Zhang, Zaber Hakim, and Chris Thomas. 2025. SteerVLM: Robust Model Control through Lightweight Activation Steering for Vision Language Models. *arXiv preprint arXiv:2510.26769* (2025). doi:10.48550/arXiv.2510.26769
- [14] Otilia Stretcu, Edward Vendrow, Kenji Hata, Krishnamurthy Viswanathan, Vittorio Ferrari, Sasan Tavakkol, Wenlei Zhou, Aditya Avinash, Emming Luo, Neil Gordon Alldrin, et al. 2023. Agile modeling: From concept to classifier in minutes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 22323–22334.
- [15] Jo Vermeulen, Kris Luyten, Elise Van Den Hoven, and Karin Coninx. 2013. Crossing the bridge over Norman’s Gulf of Execution: revealing feedforward’s true identity. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 1931–1940.
- [16] Vladimir Zaigrajew, Hubert Baniecki, and Przemyslaw Biecek. 2025. Interpreting CLIP with hierarchical sparse autoencoders. *arXiv preprint arXiv:2502.20578* (2025).
- [17] Kaiyang Zhou, Jingkan Yang, Chen Change Loy, and Ziwei Liu. 2022. Conditional prompt learning for vision-language models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 16816–16825.
- [18] Kaiyang Zhou, Jingkan Yang, Chen Change Loy, and Ziwei Liu. 2022. Learning to prompt for vision-language models. *International journal of computer vision* 130, 9 (2022), 2337–2348.

A UI and LLM Descriptor Details for User Study

Each query page showed one fixed Baseline gallery (5 images), one Linear gallery (5 images), editable positive/negative attribute fields, and global α/β sliders before applying refinement. After each query, participants also recorded method preference and Likert ratings; interaction logs and annotations were stored in Firestore.

The Linear Steering condition was run as a 3-turn, memory-aware interaction: after each turn, participants could revise critiques, and the system carried forward confirmed attributes from earlier turns to form the next steered query; after at most 3 rounds, participants pressed *Satisfied* to lock that query and continue.

For descriptor initialization, we used Groq chat completions with model llama-3.3-70b-versatile (temperature 0.3) and a fixed prompt template. The core instructions were:

"You are a vision-language retrieval expert specialising in CLIP embedding steering. You decompose queries into OBSERVABLE VISUAL attributes—body language, textures, colours, postures, spatial arrangements—never abstract or subjective words. Follow the few-shot examples closely. Return ONLY valid JSON."

You are an expert at decomposing image-search queries into visually grounded CLIP steering attributes.

CONTEXT

- We steer a CLIP ViT-B/16 query embedding by adding positive attribute embeddings and subtracting negative ones.
- Attributes must be OBSERVABLE VISUAL properties things you can literally SEE in a photograph.
- Each attribute is 1-5 words. No abstract concepts. No full sentences.

CRITICAL RULES FOR GOOD ATTRIBUTES

- For EMOTIONS or SUBJECTIVE states: describe the BODY LANGUAGE, not the emotion word.
BAD: "guilty expression" (CLIP doesn't understand abstract emotions well)
GOOD: "looking down", "avoiding eye contact", "head lowered", "droopy ears"
- For PHYSICAL descriptions: use concrete visual details.
BAD: "beautiful dog" (vague, subjective)
GOOD: "light golden fur", "floppy ears", "broad head", "medium to long wavy coat"
- For SCENES: describe observable elements.
BAD: "exciting scene" (abstract)
GOOD: "jumping", "running", "movement", "sports"
- NEGATIVES must be the visual OPPOSITE or a common CLIP confusion.
They should suppress what CLIP tends to retrieve incorrectly for this query.

FEW-SHOT EXAMPLES (from human experts):

Query: "a dog looking guilty"

positive: ["sad", "droopy", "ashamed", "looking down", "submissive", "avoiding eye contact", "head down"]
negative: ["happy", "playful", "energetic", "excited", "proud", "confident"]

Query: "a nervous-looking dog"

positive: ["anxious", "scared", "worried", "trembling", "wide eyes", "ears back"]
negative: ["confident", "relaxed", "happy", "calm", "bold"]

Query: "a golden retriever"

positive: ["golden retriever dog", "light golden fur", "medium to long wavy coat", "floppy ears", "broad head"]
negative: ["short fur dog", "dark colored fur", "pointed upright ears", "small dog breed"]

Query: "a person looking guilty"

positive: ["worried", "nervous", "serious", "sad", "secretive", "uncomfortable"]

negative: ["smiling", "happy", "confident", "laughing", "relaxed"]

Query: "an exciting action scene"

positive: ["sports", "jumping", "running", "movement"]
negative: ["standing", "sitting", "calm", "still"]

Query: "a friendly-looking dog"

positive: ["happy", "playful", "gentle", "cute", "adorable", "wagging tail"]
negative: ["aggressive", "scary", "angry", "mean", "threatening"]

NOW GENERATE FOR THIS QUERY:

USER QUERY: "{query}"

INSTRUCTIONS:

1. Generate 5-8 POSITIVE attributes (observable visual cues that define what the user wants).
 - The first 1-2 should be the most important (weight 0.8-1.0).
 - The rest are supporting visual details (weight 0.4-0.7).
 - For subjective/emotional queries: describe body language, posture, facial features.
2. Generate 5-8 NEGATIVE attributes (visual opposites or CLIP confusions to suppress).
 - Weight 0.5-0.8 for strong opposites, 0.3-0.5 for subtle.
3. Set $\alpha=0.4$ and $\beta=0.4$ (safe defaults). Only increase to 0.45 for very specific queries.

Return ONLY a JSON object (no markdown, no explanation)

```
{
  "positive": [
    {"attribute": "short visual phrase", "weight": 0.9, "rationale": "why"},
    {"attribute": "another phrase", "weight": 0.6, "rationale": "why"}
  ],
  "negative": [
    {"attribute": "short visual phrase", "weight": 0.7, "rationale": "why"},
    {"attribute": "another phrase", "weight": 0.5, "rationale": "why"}
  ],
  "alpha": 0.4,
  "beta": 0.4
}
```

The template used a fixed system instruction and few-shot user prompt to enforce *observable visual* phrasing (e.g., posture, gaze, texture, color, spatial layout) and avoid abstract labels. It requested 5-8 positive attributes and 5-8 negative attributes, each with a weight, plus default $\alpha = 0.4$ and $\beta = 0.4$. The expected response format was JSON with fields positive, negative, alpha, and beta; each attribute entry contained attribute, weight, and optional rationale text.

B Pipeline Transparency and Robustness

Attribute extraction pipeline. A constrained few-shot LLM prompt generates structured JSON descriptors containing positive and negative attribute phrases, per-attribute weights, and global α/β values. Outputs are validated, non-list fields are replaced with empty lists, and numeric values are clamped to safe ranges before use. Each attribute phrase is encoded with the frozen CLIP text encoder as a photo of {attribute} and steered via Equation (1). The full descriptor list and parameter values are shown to participants after each round. In user refinement rounds, participants directly edit

the positive and negative lists, which replace the LLM descriptors for the next retrieval pass.

Robustness and failure handling. The pipeline handles failures at multiple levels. If the LLM client is unavailable, the system falls back to a curated query-specific attribute bank. Any runtime or parse exception is caught and routed to the same fallback. Malformed outputs are sanitized before use. Beyond automated handling, participants can directly edit incorrect attributes and re-run retrieval, correcting extraction errors during the session. All applied attributes and retrieved results are logged to Firestore, enabling post-hoc analysis of which descriptors were active across rounds.